

Architecture Justification

Architecture Selection: We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

Architectural Choice

The kiosk is organised into three horizontal layers:

- **Presentation Layer:** User Interface Component (touch screen)
- **Business Layer:** Order Manager, Payment Service and Receipt Printer components
- **Data Layer:** Database Component (menu and pricing)

Each layer depends only on the layer below it, through well-defined interfaces (Order, Payment, Receipt and Database Interfaces).

Reasons for the Choice

1. **The scope is small and fixed.** The kiosk offers only 3 coffee types, 2 sizes and card-only payment. Its flow is linear: select, pay, print. This maps directly onto a layered structure. Microservices would add deployment and networking overhead that a single kiosk does not need, and Client-Server would be unnecessary since the kiosk runs standalone.
2. **Clear separation of concerns and replaceability.** Touch-screen handling, order logic, the payment gateway, printer hardware and menu storage are each isolated in their own component. Because components interact only through interfaces, we can swap the printer model or payment provider, or add a new coffee type or size, by changing one component without affecting the others.

Security Advantage

All payment handling is confined to the Payment Service Component, and the UI never accesses it directly. It can only reach payment functions through the Order Manager. This limits the attack surface, keeps card data out of the presentation layer and database, and allows security controls (encryption, tokenisation, input validation) to be enforced at a single point. This supports PCI-DSS compliance, which matters since credit card is the only payment method.

Performance Benefit

Everything runs locally on the kiosk, so calls between layers are simple in-process calls with no network latency, unlike microservices. Menu and pricing data can be cached in the Business Layer after being read from the database once. This gives the customer fast touch-screen responses and quick order processing, even during busy café hours.